



What Is Enterprise MLOps?

From Experiments to Governed, Scalable AI Products

A framework for productionizing machine learning at scale while maintaining governance, reliability, and security

Why Enterprise MLOps Matters



Reliable Services

Transform experimental notebooks into robust, repeatable production services



Faster Delivery

Dramatically reduce time-to-production and operational risk



Aligned Lifecycles

Synchronize data, model, and software development workflows



Enterprise Controls

Meet security, compliance, and cost constraints at scale

Core Pillars of Enterprise MLOps



1

Platform

Compute, storage, networking infrastructure and orchestration systems

2

Pipeline

End-to-end workflow from data ingestion through feature engineering, training, evaluation, and deployment

3

Operations

Monitoring, alerting, and incident response protocols

4

Governance

Lineage tracking, approval workflows, audit capabilities, and risk controls

Enterprise MLOps Reference Architecture



Data Layer

Lakes/warehouses, catalogs, feature stores with governed access



ML Layer

Experiment tracking, model registry, CI/CD for ML assets



Serving Layer

Batch/real-time endpoints, A/B testing, canary deployments



Observability

Model/data drift, quality metrics, cost tracking, SLOs



Security

IAM, secrets management, policy-as-code enforcement

ML Lifecycle Stages

Enterprise MLOps enables a consistent, repeatable process across all AI initiatives

Problem & Data Discovery

Define business problem, identify data sources, assess feasibility

Monitoring & Feedback

Track performance, detect drift, trigger retraining loops

Deployment

Release via batch, streaming, or online serving patterns



Feature Engineering

Create features with offline/online parity, ensure reproducibility

Training & Tracking

Run experiments, version code/data/models, capture metrics

Validation & Approval

Execute tests, conduct bias/safety checks, obtain approvals

Key Platform Capabilities

The foundation for scalable, enterprise-grade ML systems



Reproducible Environments

Containerized workloads with infrastructure-as-code ensuring consistency across development, testing, and production



Standardized Pipelines

Orchestration using Airflow, Kubeflow, Vertex, or SageMaker with reusable components



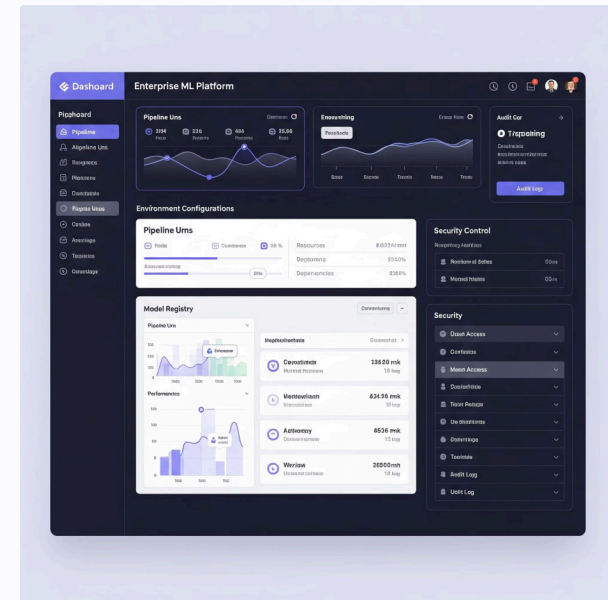
Model Registry + Promotion

Centralized catalog with versioning and stage-gated promotion workflows



Secrets & Access Control

Managed secrets with least-privilege access controls for all ML assets



Data & Feature Management



Feature Store Benefits

- **Consistency** between offline training and online serving
- **Reuse** across projects to avoid duplication of effort
- **Data contracts** with enforced schemas prevent errors
- **Quality checks** for missingness, skew, and leakage
- **Lineage tracking** from raw data to features to models
- **Point-in-time correctness** prevents training-serving skew

Deployment Patterns

Choose the right serving pattern based on latency requirements and data volumes

Batch Scoring

- Scheduled ETL/ELT jobs
- High throughput
- Minutes-to-hours latency
- Efficient for large volumes

Streaming

- Kafka, Flink-based pipelines
- Near-real-time processing
- Seconds-to-minutes latency
- Event-driven architecture

Online Serving

- FastAPI/Triton microservices
- Autoscaling capabilities
- Milliseconds-to-seconds latency
- Safe rollout patterns

Observability & SLOs

Comprehensive monitoring across all dimensions of ML systems

Model Health

- Accuracy, precision, recall, F1
- Calibration quality
- Concept/data drift detection
- Fairness across segments

System Health

- p95/p99 latency tracking
- Throughput measurements
- Error rates and types
- Resource utilization

Data Health

- Distribution shifts over time
- Data freshness metrics
- Missing values and outliers
- Schema enforcement

Business KPIs

- Conversion rates
- Risk assessments
- Cost per prediction
- Business value delivered



Governance & Risk Management

Policy Enforcement

Automated guardrails that scale with your ML operations



Policy-as-Code

Programmatic approval workflows and promotion gates



Audit Trails

Complete data/model/prompt lineage for compliance



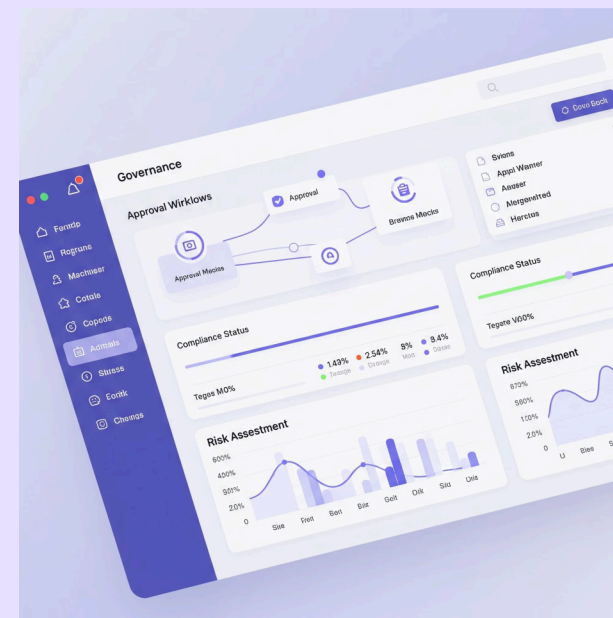
PII Handling

Defined retention, deletion, and anonymization workflows



Safety Evaluations

Bias, robustness, and safety checks as deployment gates





Organization & Process

Roles & Responsibilities

Clear RACI matrix across data engineering, ML engineering, platform, and security teams

- Who builds models vs. infrastructure
- Who approves promotions
- Who responds to incidents

ML-Specific CI/CD

Specialized pipelines that handle model artifacts and data dependencies

- Unit + data tests
- Evaluation tests
- Integration tests

Templates & Playbooks

Standardized approaches for common ML project types

- Project bootstrapping
- Technology selection guides
- Documentation standards

FinOps Practices

Cost management across the ML lifecycle

- \$/experiment tracking
- \$/inference monitoring
- Resource optimization

What "Good" Looks Like

Measurable outcomes of a mature enterprise MLOps practice

Days

From Idea to
Production

Not months or quarters -
automated pipelines and
self-service capabilities
accelerate time-to-value

100%

Reproducibility

Every deployment can be
recreated precisely with
versioned artifacts and
documented dependencies

0

Manual Approvals

Automated tests, checks and
policies replace human
gatekeepers for standard
workflows

24/7

Monitoring
Coverage

Continuous monitoring with
drift detection triggers
automated retraining where
safe and appropriate

"Self-serve platform with guardrails, not gatekeepers"

Key Takeaways

Enterprise MLOps is Comprehensive

It combines technology, process, and governance into a cohesive framework

Standardization is Critical

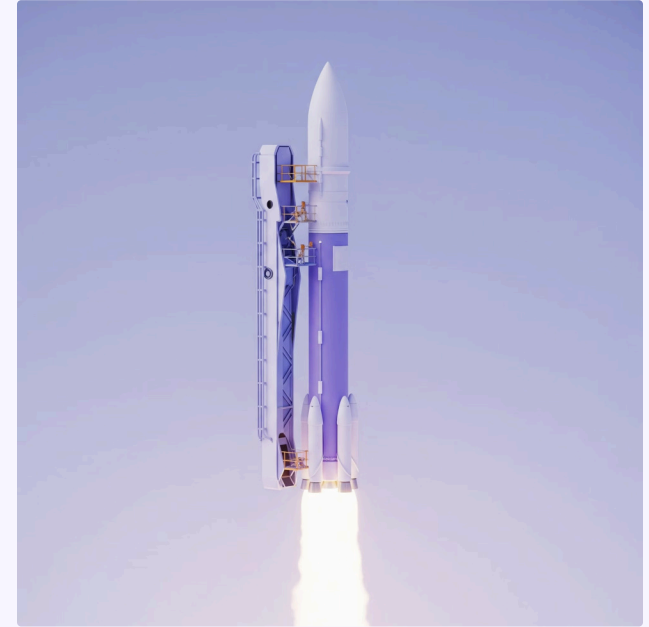
Standardize pipelines, centralize features, and track everything for consistency

Safety First

Ship confidently with controlled rollouts, SLOs, and compliance controls

Speed with Guardrails

Optimize for velocity with automated guardrails, not ad-hoc heroics



Questions?